

Emotion classification in text: a comparative study of a Bayesian classifier and a transformer model

Sourabh Tanwar¹, Saksham Yash Srivastava^{1*}

¹ Department of Electronics and Communication Engineering, Bharati Vidyapeeth, Pune, India

* Corresponding author: [sourabh24-ece@bvucoep.edu.in
sysrivastava24-ece@bvucoep.edu.in]

Abstract

Text emotion classification is an important natural language processing task with applications in sentiment analysis, conversational systems, social media monitoring, and human-computer interaction. This study presents a comparative evaluation of a traditional Bayesian approach and a modern transformer-based approach for classifying emotions expressed in text. The *dair-ai/emotion* dataset containing 20,000 text samples across six emotion categories—sadness, joy, love, anger, fear, and surprise—is used for evaluation. The classical approach employs Term Frequency–Inverse Document Frequency (TF–IDF) feature extraction followed by a Multinomial Naive Bayes classifier, while the modern approach uses a fine-tuned DistilBERT transformer model. On the test set, Multinomial Naive Bayes achieved an accuracy of 69.40% and a macro F1-score of 44.00%, whereas the DistilBERT model achieved an accuracy of 92.30% and a macro F1-score of 87.70%. The results demonstrate the advantage of context-aware transformer representations over traditional word-frequency-based features for this emotion classification task.

Keywords: Bayesian Classifier; DistilBERT; Emotion Classification; Multinomial Naive Bayes; Natural Language Processing; Text Classification; TF–IDF; Transformer

Introduction

Emotion classification is a text classification problem in which a system assigns an emotion label to an input sentence or document. It is an important natural language processing task because emotional information can complement semantic information extracted from text. Applications include conversational agents, social-media analysis, customer feedback processing, and human-computer interaction.

Classical text classification systems commonly represent documents using sparse word-based features and apply probabilistic or linear classifiers, while modern systems increasingly use contextual neural representations.

TF–IDF is a widely used representation for transforming text into numerical features by increasing the importance of terms that are frequent in a document but relatively uncommon across a collection [1]. Multinomial Naive Bayes is an established baseline for text classification because it is computationally efficient and can operate with high-dimensional sparse representations [2, 3]. However, its conditional independence assumption limits its ability to represent relationships among words and broader sentence context.

The Transformer architecture introduced self-attention for modeling relationships among tokens in a sequence [4]. BERT demonstrated the effectiveness of bidirectional

contextual pretraining for language understanding [5], while DistilBERT provided a smaller distilled alternative with reduced computational requirements [6]. Modern implementations are also accessible through the Hugging Face Transformers framework [7].

The present study compares Multinomial Naive Bayes with a fine-tuned DistilBERT emotion classifier on the same six-class benchmark. The study addresses three practical questions: how strong is a simple TF-IDF plus Naive Bayes baseline, how much performance improvement is obtained from a contextual transformer, and how do the two approaches behave across individual emotion classes?

Macro and weighted F1 scores are reported in addition to accuracy to account for class imbalance.

Materials and methods

Dataset

The experiments use the `dair-ai/emotion` dataset, an English emotion classification benchmark containing six emotion categories: sadness, joy, love, anger, fear, and surprise [9].

The dataset provides separate training, validation, and test splits. The training split contains 16,000 samples, the validation split contains 2,000 samples, and the test split contains 2,000 samples, giving 20,000 samples in total.

The training distribution is imbalanced. It contains 5,362 joy samples, 4,666 sadness samples, 2,159 anger samples, 1,937 fear samples, 1,304 love samples, and 572 surprise samples.

Therefore, joy and sadness are the most frequent classes, whereas surprise is substantially less frequent. This imbalance is important when interpreting accuracy because a model can obtain reasonable overall accuracy while performing poorly on minority classes.

The classification task can be represented as learning a function

$$f : X \rightarrow Y, \quad (1)$$

where X represents an input text and Y is one of the six emotion labels. For an unseen sentence x , the system produces a predicted label

$$\hat{y} = f(x). \quad (2)$$

The training-set distribution is shown in Fig 1. The figure highlights the class imbalance used later to interpret macro and weighted evaluation measures.

Text preprocessing and TF-IDF representation

For the Naive Bayes pipeline, input text is converted to lowercase and common English stop words are removed. The resulting documents are transformed into a sparse TF-IDF matrix.

The vectorizer is fitted only on the training split, and the learned vocabulary and inverse-document-frequency values are reused when transforming the test set. This prevents information from the test set from being used to construct the training representation.

For a term t in document d , TF-IDF can be represented conceptually as

$$\text{TFIDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t), \quad (3)$$

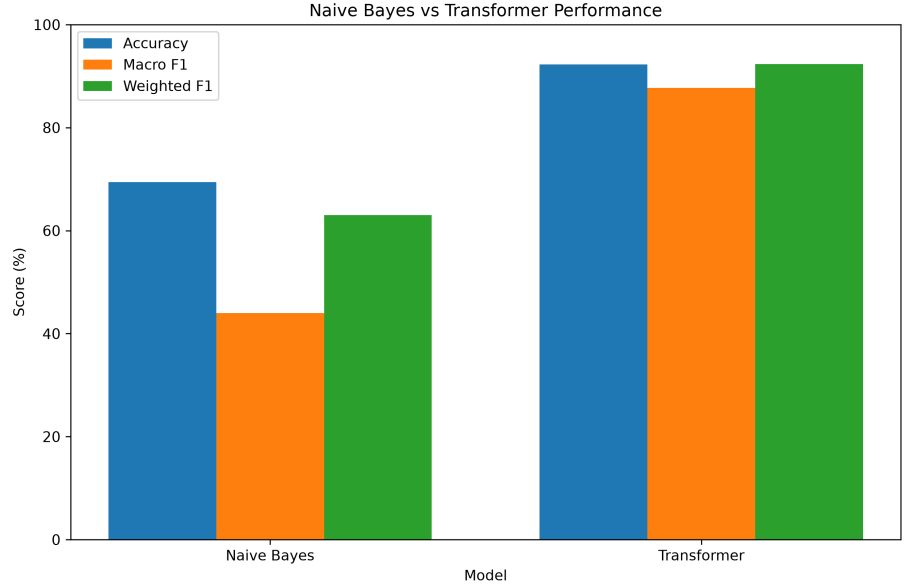


Fig 1. Emotion distribution in the training dataset. The six emotion classes are shown with their corresponding numbers of training samples.

where TF measures the occurrence of a term in a document and IDF reduces the influence of terms appearing across many documents [1].

Multinomial Naive Bayes

The first classifier is Multinomial Naive Bayes, a probabilistic model commonly used for text classification [2].

Given a document representation x and class c , the classifier selects the class with the highest posterior probability:

$$\hat{c} = \arg \max_c P(c | x). \quad (4)$$

Using Bayes' rule, the posterior is proportional to the likelihood of the observed features under the class multiplied by the class prior.

The model assumes conditional independence among features given the class. Although this assumption is simplified, it provides a fast and interpretable baseline for sparse text representations [3].

In this implementation, TF-IDF features were generated using the training text, and a Multinomial Naive Bayes classifier was fitted to the corresponding labels. The trained vectorizer and classifier were then applied to the test set without refitting.

Pseudocode 1: TF-IDF and Multinomial Naive Bayes Classification

1. Load the *dair-ai/emotion* dataset.
2. Extract the training texts and corresponding emotion labels.
3. Extract the test texts and corresponding emotion labels.
4. Convert the training text to lowercase.
5. Remove common English stop words.
6. Initialize the TF-IDF vectorizer.

7. Fit the TF-IDF vectorizer using only the training text. 82
8. Transform the training and test text into TF-IDF feature vectors. 83
9. Initialize the Multinomial Naive Bayes classifier. 84
10. Train the classifier using the TF-IDF training features. 85
11. Predict the emotion labels of the test samples. 86
12. Calculate accuracy, precision, recall, and F1-score. 87
13. Generate the confusion matrix. 88
14. Return the predicted labels and evaluation results. 89

Transformer-based model 91

The second approach uses `Sreekant13/distilbert-emotion`, a publicly fine-tuned DistilBERT checkpoint whose output classes match the six labels of the selected benchmark. 92-94

DistilBERT is derived from BERT through knowledge distillation and is designed to reduce model size and inference cost while retaining contextual language representations [6]. 95-97

BERT itself uses bidirectional Transformer representations to model contextual relationships between tokens [5]. 98-99

The model receives the original text through its associated tokenizer. Long inputs are truncated to the model's supported sequence length. 100-101

Unlike the Naive Bayes pipeline, the transformer does not use TF-IDF features; it operates on tokenized input and produces a class label with an associated prediction score. 102-104

The Transformer implementation used in the project is provided through the Hugging Face Transformers framework [7]. 105-106

An important methodological distinction is that the transformer is a publicly fine-tuned checkpoint, whereas the Naive Bayes classifier is trained directly within the project. 107-109

Therefore, the comparison evaluates practical performance between a classical baseline and an available transfer-learning model rather than equal-cost training from scratch. 110-112

Pseudocode 2: Transformer-Based DistilBERT Classification 113-114

1. Load the fine-tuned DistilBERT emotion classification model. 115
2. Initialize the tokenizer associated with the model. 116
3. Load the test text samples. 117
4. For each test sentence, tokenize the input text. 118
5. Apply truncation when the input exceeds the maximum supported sequence length. 119-120
6. Pass the tokenized input through the DistilBERT model. 121
7. Obtain the predicted emotion label. 122
8. Obtain the prediction confidence score. 123
9. Store the predicted label and confidence score. 124
10. Compare the predicted labels with the actual test labels. 125
11. Calculate accuracy, precision, recall, and F1-score. 126

12. Generate the confusion matrix. 127
13. Return the predictions, confidence scores, and evaluation results. 128

Experimental setup 129

Both approaches were evaluated on the same 2,000-sample test split. The Naive Bayes model was trained using the 16,000-sample training split. 130

The validation split was retained as part of the benchmark dataset but was not used as an additional training set in the reported Naive Bayes experiment. 131

The transformer checkpoint was loaded for inference using the same six-class label space. 132

The project was implemented in Python using scikit-learn for TF-IDF and Naive Bayes and the Transformers framework for transformer inference. Scikit-learn is a widely used Python machine-learning library [8]. 133

Evaluation metrics 140

The models were evaluated using accuracy, precision, recall, F1-score, macro F1, weighted F1, and confusion matrices. 141

Accuracy is the proportion of correctly classified test samples: 142

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \quad (5) \quad 143$$

For multiclass classification, class-wise precision, recall, and F1 are calculated and then summarized using macro and weighted averaging. 144

Macro F1 gives equal importance to each class, while weighted F1 weights each class according to its number of test samples. 145

The confusion matrix provides the number of predictions for each actual-predicted class pair. 146

Results 150

The overall performance comparison is summarized in Table 1. Both models were evaluated on exactly the same 2,000 test samples, allowing their aggregate results to be compared directly. 151

Table 1. Overall performance of the two emotion classification approaches.

Accuracy, macro F1, and weighted F1 are reported on the same 2,000-sample test set. 152

Model	Accuracy	Macro F1	Weighted F1
Multinomial Naive Bayes	69.40%	44.00%	63.00%
DistilBERT Transformer	92.30%	87.70%	92.33%

Naive Bayes results 154

The Multinomial Naive Bayes classifier achieved 69.40% accuracy. 155

Its strongest results were obtained for the frequent joy and sadness classes. The model correctly identified most sadness examples and almost all joy examples, but it showed very weak recall for love and surprise. 156

In the reported confusion matrix, no surprise examples were correctly classified. This behavior is consistent with the class imbalance observed in the training data.

The Naive Bayes confusion matrix contains 1,388 correct predictions out of 2,000 test samples.

The macro F1 score of 44.00% is considerably lower than the weighted F1 score of 63.00%, demonstrating that performance is uneven across classes.

The class-wise prediction counts are provided in Fig 2.

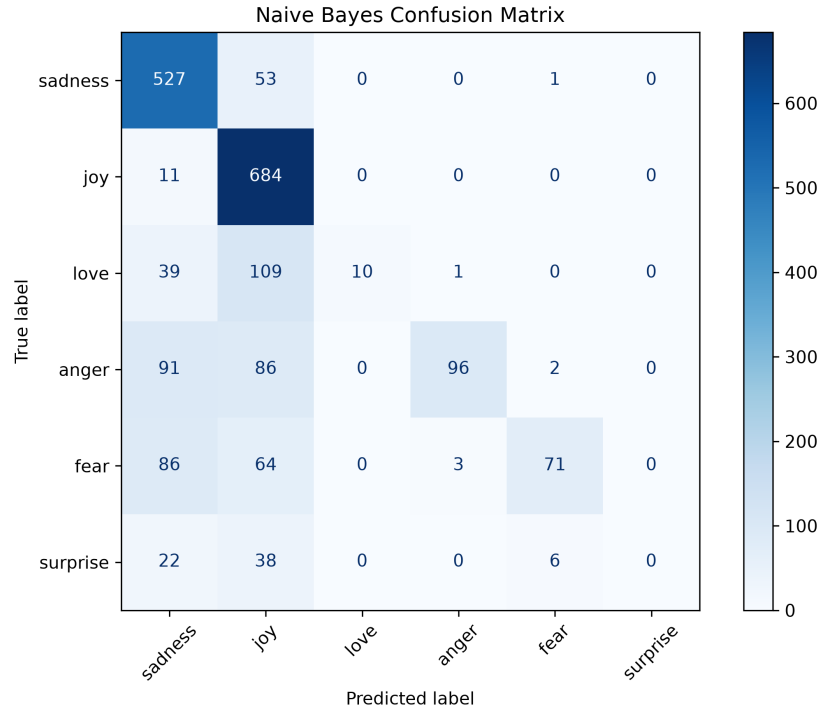


Fig 2. Confusion matrix of the Multinomial Naive Bayes classifier. The matrix shows the classification performance of the TF-IDF and Multinomial Naive Bayes approach across the six emotion categories.

Transformer results

The DistilBERT transformer achieved 92.30% accuracy, corresponding to 1,846 correct predictions out of 2,000 test samples.

The macro F1 score was 87.70% and the weighted F1 score was 92.33%.

These results show that the transformer maintained high performance across both frequent and less frequent emotion classes.

The largest observed class-level confusion involved joy and love.

Thirty-five love examples were classified as joy, while 25 joy examples were classified as love.

The transformer correctly classified 54 of the 66 surprise examples, substantially improving the minority-class result compared with Naive Bayes.

The complete confusion matrix is shown in Fig 3.

Error analysis

The error patterns of the two models differ substantially.

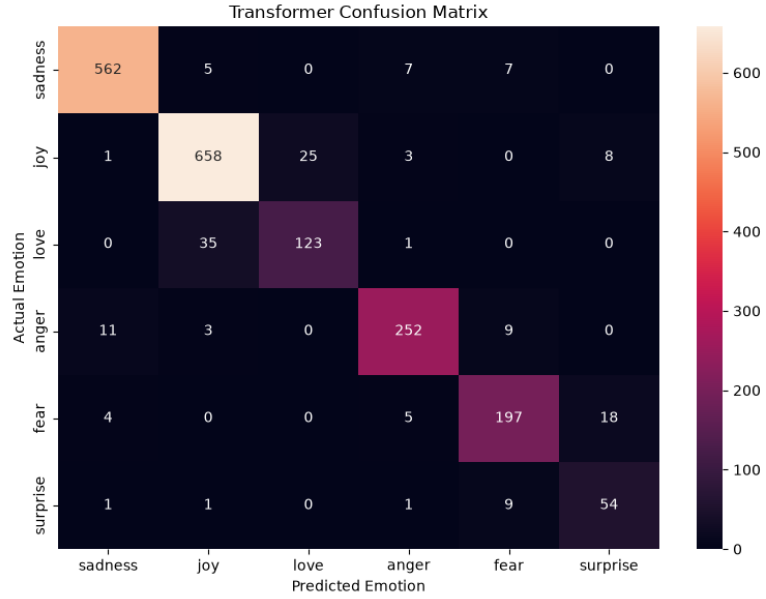


Fig 3. Confusion matrix of the transformer model. The matrix shows the classification performance of the DistilBERT-based transformer model across the six emotion categories.

Naive Bayes relies on sparse lexical evidence and therefore tends to favor classes associated with frequent words in the training data. 180

Its low recall for love and surprise is particularly important because these classes contain fewer training examples. 181

The transformer uses contextual representations and can model interactions among tokens, which helps it distinguish semantically related emotions. 182

The two approaches can still disagree on an individual sentence. 183

Such disagreement is expected because they use different representations and modeling assumptions. 184

The transformer should not be considered automatically correct for every sentence; its reported advantage is based on aggregate performance on the labeled test set. 185

Model comparison 186

The transformer improved accuracy by 22.90 percentage points compared with Naive Bayes. 187

The macro F1 improvement was 43.70 percentage points, indicating that the gain was particularly important for balanced class-level performance. 188

The weighted F1 improvement was 29.33 percentage points. 189

These differences indicate that contextual modeling provides a substantial advantage for this benchmark under the experimental conditions used in this study. 190

Discussion 191

The experimental results demonstrate a clear performance difference between the traditional and transformer approaches. 192

The accuracy and macro F1 improvements indicate that contextual representations are particularly valuable for a six-class emotion task with substantial class imbalance. 193

The difference between macro and weighted F1 for Naive Bayes also shows why accuracy alone is insufficient for this experiment.

The comparison should be interpreted carefully.

The Naive Bayes model was trained directly on the project training split, whereas the DistilBERT model was already fine-tuned for the same benchmark task.

Consequently, the study demonstrates the practical performance of a classical baseline against an available transfer-learning model, rather than comparing two models trained from scratch under identical computational budgets.

The selected dataset is also narrower than broader emotion resources such as GoEmotions, which contains a larger set of fine-grained emotion categories and was developed from Reddit comments [11].

Therefore, the reported performance should not be interpreted as general performance for every emotion taxonomy or text domain.

Importance of contextual information

One of the main differences between the two approaches is the information available to the classifier.

TF-IDF captures statistical relationships between words and documents. It is effective at identifying informative vocabulary but does not naturally represent the complete contextual meaning of a sentence.

Transformer representations are contextual.

The representation of a token can depend on other tokens in the sequence.

This is particularly useful when the same word can appear in multiple emotional contexts.

For example, a word associated with a negative event does not necessarily indicate sadness in every sentence. The surrounding language may instead indicate fear, anger, surprise, or another category.

Contextual modelling can therefore provide a more flexible representation.

Effect of class imbalance

The distribution of samples is not uniform.

Joy and sadness contain many more samples than surprise.

If a model performs particularly well on common categories but poorly on rare categories, its overall accuracy may remain relatively high.

This is why macro F1-score is important for interpreting the present results.

The macro F1-score treats every category equally regardless of the number of samples.

The large difference between the Naive Bayes macro and weighted F1-scores suggests that the classical model's performance varies considerably between categories.

In contrast, the smaller difference observed for the transformer suggests more consistent performance.

Classical baseline versus pretrained model

A classical baseline is still important in machine learning experiments.

Without a baseline, it is difficult to determine whether a more complex model provides a meaningful improvement.

The Naive Bayes classifier provides a computationally simple reference point.

Its relatively low complexity makes it easy to train, evaluate, reproduce, and deploy.

The transformer model represents a more advanced approach.

Its stronger performance demonstrates the practical value of contextual language representations for the target classification task.
Therefore, the results do not imply that Naive Bayes is universally inferior.
Instead, they demonstrate that for this particular emotion classification task and the reported evaluation setup, the transformer provides substantially stronger predictive performance.

Error analysis

Error analysis is an important component of multi-class classification because aggregate metrics alone do not reveal why individual predictions fail.
In emotion classification, errors may arise from lexical similarity, ambiguous language, short messages, implicit emotions, and overlap between categories.
The Naive Bayes model's errors are particularly visible for minority classes. Its confusion matrix shows that the classifier frequently assigns minority examples to more frequent categories.
The transformer reduces many of these errors by using contextual representations.
Nevertheless, some confusion remains between semantically related categories, particularly joy and love.

Limitations

Several limitations should be considered when interpreting the results.
First, the transformer model used in this study is a publicly fine-tuned checkpoint rather than a model trained from scratch on the project training split.
Second, the two approaches have different computational requirements.
Naive Bayes is lightweight and can be trained rapidly on CPU-based systems, whereas transformer inference generally requires greater memory and processing resources.
Third, the dataset contains class imbalance. The surprise class is much smaller than joy and sadness, which affects the stability of class-level metrics.
Fourth, the experiment focuses on English text and therefore does not directly establish performance for multilingual or code-mixed text.
Finally, the benchmark consists of relatively short text samples. Performance may differ on longer documents, conversational transcripts, or domain-specific text.

Ethical considerations

Emotion classification systems should be interpreted as predictive tools rather than definitive measurements of a person's internal emotional state.
Text can contain ambiguity, sarcasm, cultural variation, or context that is not available to the classifier.
Therefore, predictions should not be used as the sole basis for sensitive decisions involving individuals.
The dataset and pretrained model should also be used according to their respective licensing and usage conditions.

Future work

Several directions can extend the present study.

First, the transformer model could be fine-tuned directly on the training portion of the emotion dataset. Such an experiment would provide a more controlled comparison between classical supervised learning and transformer-based supervised learning.

Second, detailed per-class precision, recall, and F1-score analysis could identify which emotion categories are most difficult to classify.

Third, data augmentation and class balancing strategies could be investigated to improve performance on less frequent emotions.

Fourth, computational benchmarking could compare training time, inference latency, memory consumption, and model size.

Fifth, domain-specific evaluation could test whether the models generalize beyond the original dataset.

Testing on social-media text, customer feedback, conversational text, or other emotion datasets would help determine the robustness of the observed performance difference.

Multilingual and code-mixed emotion classification is another potential direction.

Explainability techniques could also be investigated to identify which words or contextual relationships contribute most strongly to individual emotion predictions.

Conclusion

This study presented a comparative evaluation of a TF-IDF based Multinomial Naive Bayes classifier and a DistilBERT-based transformer model for six-class emotion classification.

The Naive Bayes classifier achieved 69.40% accuracy, 44.00% macro F1-score, and 63.00% weighted F1-score.

The transformer model achieved 92.30% accuracy, 87.70% macro F1-score, and 92.33% weighted F1-score.

The transformer therefore improved accuracy by 22.90 percentage points and macro F1-score by 43.70 percentage points compared with the classical baseline.

The results demonstrate that contextual transformer representations provide substantially stronger classification performance than the traditional TF-IDF and Naive Bayes approach for the evaluated emotion classification task.

The improvement is particularly significant for macro F1-score, indicating more consistent performance across the six emotion categories.

At the same time, the Naive Bayes approach remains valuable because of its simplicity, speed, interpretability, and lower computational requirements.

This makes it useful for baseline experiments, resource-constrained applications, and situations where model simplicity is important.

The comparison therefore illustrates an important machine learning trade-off.

Traditional methods can provide efficient and understandable solutions, while transformer models can provide significantly stronger contextual language understanding at the cost of greater computational complexity.

Overall, the findings support the use of contextual transformer models when high predictive performance is the primary objective, while classical methods remain relevant when computational efficiency and simplicity are major considerations.

Supporting information

S1 Fig. Training dataset distribution.

The supporting image contains the six-class training distribution used for the dataset analysis.

S2 Fig. Model confusion matrices.

The supporting images contain the Naive Bayes and DistilBERT confusion matrices evaluated on the 2,000-sample test set.

S1 File. Project source code and documentation.

The source code, requirements, README, and execution instructions are maintained with the accompanying project materials.

Acknowledgments

The authors would like to acknowledge the guidance and support received during the development of this study.

The authors also acknowledge the publicly available emotion dataset and open-source natural language processing tools used in this work.

References

1. Manning CD, Raghavan P, Schütze H. Introduction to Information Retrieval. Cambridge: Cambridge University Press; 2008.
2. McCallum A, Nigam K. A comparison of event models for naive Bayes text classification. In: AAAI-98 Workshop on Learning for Text Categorization; 1998. p. 41–48.
3. Rennie JDM, Shih L, Teevan J, Karger DR. Tackling the poor assumptions of naive Bayes text classifiers. Proceedings of the 20th International Conference on Machine Learning. 2003. p. 616–623.
4. Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser L, Polosukhin I. Attention is all you need. Advances in Neural Information Processing Systems. 2017;30.
5. Devlin J, Chang MW, Lee K, Toutanova K. BERT: Pre-training of deep bidirectional transformers for language understanding. Proceedings of NAACL-HLT. 2019. p. 4171–4186.
6. Sanh V, Debut L, Chaumond J, Wolf T. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. arXiv. 2019;1910.01108.
7. Wolf T, Debut L, Sanh V, Chaumond J, Delangue C, Moi A, Cistac P, Rault T, Louf R, Funtowicz M, Davison J, Shleifer S, von Platen P, Ma C, Jernite Y, Plu J, Xu C, Le Scao T, Gugger S, Drame M, Lhoest Q, Rush AM. Transformers: State-of-the-art natural language processing. Proceedings of EMNLP: System Demonstrations. 2020. p. 38–45.
8. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, Blondel M, Prettenhofer P, Weiss R, Dubourg V, Vanderplas J, Passos A, Cournapeau D, Brucher M, Perrot M, Duchesnay E. Scikit-learn: Machine learning in Python. Journal of Machine Learning Research. 2011;12:2825–2830.
9. dair-ai. Emotion dataset. Hugging Face Datasets. Available from: Hugging Face dataset repository.
10. Sreekant13. DistilBERT emotion classifier. Hugging Face Model Hub. Available from: Hugging Face model repository.

11. Demszky D, Movshovitz-Attias D, Ko J, Cowen A, Nemade G, Ravi S. GoEmotions: A dataset of fine-grained emotions. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics. 2020. p. 4040–4054.
12. Mikolov T, Chen K, Corrado G, Dean J. Efficient estimation of word representations in vector space. arXiv. 2013;1301.3781.
13. Joulin A, Grave E, Bojanowski P, Mikolov T. Bag of tricks for efficient text classification. Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics. 2017;2:427–431.
14. Liu Y, Ott M, Goyal N, Du J, Joshi M, Chen D, Levy O, Lewis M, Zettlemoyer L, Stoyanov V. RoBERTa: A robustly optimized BERT pretraining approach. arXiv. 2019;1907.11692.

Appendix

A. Project Source Code and Reproducibility

The complete source code, project documentation, requirements file, and supporting project materials for the emotion classification system are available in the following GitHub repository:

<https://github.com/sakshamys/emotion-classification-bayes-vs-transformer>

The repository contains the implementation of the TF-IDF based Multinomial Naive Bayes classifier and the transformer-based DistilBERT classifier used in this study. It also provides the project documentation and instructions required to reproduce the experimental setup.

B. Dataset and Model Information

The project uses the `dair-ai/emotion` dataset containing six emotion classes: sadness, joy, love, anger, fear, and surprise.

The classical classification pipeline consists of TF-IDF feature extraction followed by Multinomial Naive Bayes classification.

The transformer pipeline uses the publicly available `Sreekant13/distilbert-emotion` checkpoint, whose output classes match the six emotion categories used in this study.

C. Experimental Results

The final test-set results reported in this manuscript are summarized below.

Model	Accuracy	Macro F1	Weighted F1
Multinomial Naive Bayes	69.40%	44.00%	63.00%
DistilBERT Transformer	92.30%	87.70%	92.33%

The transformer therefore achieved an accuracy improvement of 22.90 percentage points over the Multinomial Naive Bayes baseline.

D. Figures

The manuscript includes the following experimental figures:

1. Training dataset distribution across the six emotion classes.
2. Confusion matrix of the Multinomial Naive Bayes classifier.
3. Confusion matrix of the DistilBERT transformer classifier.

The corresponding figure files are included with the manuscript source and are also available with the project materials in the GitHub repository.

E. Pseudocode

The manuscript provides pseudocode for both major classification pipelines:

1. TF-IDF feature extraction and Multinomial Naive Bayes classification.
2. Tokenization and DistilBERT-based transformer classification.

These pseudocode descriptions summarize the main computational steps used during the experiments.